

**Quiz No. 1****CS3004****Software Design and Analysis****Total marks: 30****Total time: 40 minutes****Date: 13/02/2026****Roll No.** _____**Name.** _____**Scenario: Online Food Delivery System**

A food delivery platform allows registered customers to order meals from multiple restaurants. Each customer has a unique customer ID, name, email, phone number, and delivery address. Restaurants listed on the platform have a unique restaurant ID, name, location, contact number, and rating. Each restaurant offers a menu consisting of multiple food items. Each food item has a name, description, price, category, and availability status. A customer places an order that may include one or more food items from a single restaurant. Each order records the order date/time, total amount, delivery status (pending, preparing, out for delivery, delivered), and payment method. Delivery riders are responsible for delivering orders. Each rider has a unique rider ID, name, phone number, and vehicle type. A rider can handle multiple deliveries per day but can only deliver one order at a time. Customers can also rate and review restaurants after receiving their orders. The system keeps track of ratings given by customers.

Question: Read the Online Food Delivery System scenario given above and answer the following questions.

1. Identify all candidate conceptual classes from the scenario above.
2. From your identified candidate conceptual classes, identify any non-classes, subtypes or redundant concepts that you would not retain as a conceptual class and justify each briefly.
3. Propose three associations that are critical for enforcing key business rules in the scenario. And for each:
 - a. Name the two conceptual classes involved.
 - b. Specify the role and multiplicity of the association.
 - c. Explain in one sentence which business rule it helps enforce.
4. Create a complete UML Domain Model Diagram that includes, at least 6 conceptual classes, attributes for 4 of them and five associations with correct multiplicities.
5. The platform now wants to introduce discount coupons that customers can apply to orders. Suggest one new conceptual class to add to the model. List its key attributes and describe one association connecting it to an existing class.